PLATFORM ENGINEERING SHOWCASE

Homelab Platform

A production-style Kubernetes platform on three small hosts at home. Everything is code, Flux applies every change from Git, and every request passes through SSO, policy and telemetry.

3
Proxmox VE hosts in one cluster

8
Debian 13 VMs, built by Terraform

3 + 5
k0s controllers (etcd) and workers

100%
GitOps: Flux owns every app

940
Mbit/s measured between every host pair

Highlights

- Public apps come in only through a Cloudflare Tunnel. Private ones stay on Tailscale or the home LAN.
- Own rate limiter and WAF (dvarapala, in Rust) runs as an operator-managed gateway in front of each public app.
- One Google SSO (Cloudflare Access as the OIDC issuer) for Proxmox, kube-apiserver, OpenBao, Headlamp, LiteLLM and n8n.
- Secrets live in OpenBao and reach pods through External Secrets. Kubernetes Secrets are encrypted in etcd.
- Kyverno admission policy, Tetragon runtime enforcement, Wazuh endpoint security and an apiserver audit log.
- WebAssembly workloads on a Wasmtime RuntimeClass next to normal containers.
- AI agents in n8n review code, open PRs and fix CI, through a LiteLLM gateway with spend tracking.
- Point-in-time Postgres recovery, OpenBao and etcd snapshots in Cloudflare R2, with tested restores.

Stack

Proxmox VE 9 · k0s · Cilium · Flux CD · Terraform · Gateway API

WireGuard · Tailscale · Cloudflare Tunnel · Cloudflare Access · OpenBao

External Secrets · cert-manager · Kyverno · Tetragon · Wazuh

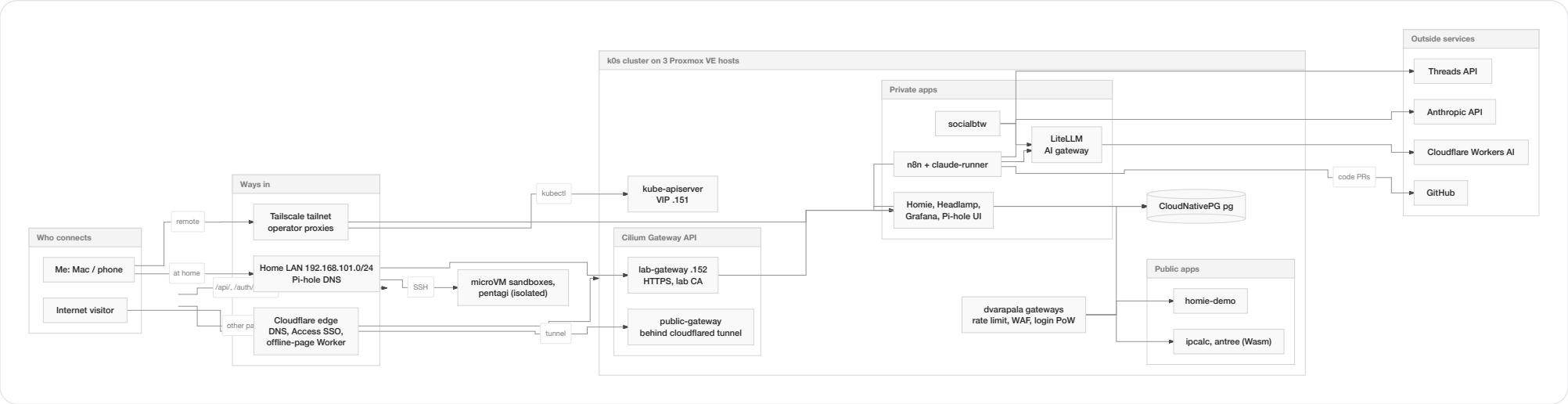
CloudNativePG · OpenTelemetry · VictoriaMetrics · VictoriaLogs

Grafana · Alloy · Wasmtime · LiteLLM · n8n · Claude Code

GitHub Actions + ARC · Renovate · Pi-hole · Rust

User traffic

Three ways in. I use Tailscale when away and the LAN at home. Internet visitors reach only namespaces labelled `exposure=public`, through the Cloudflare Tunnel and a dvarapala gateway.



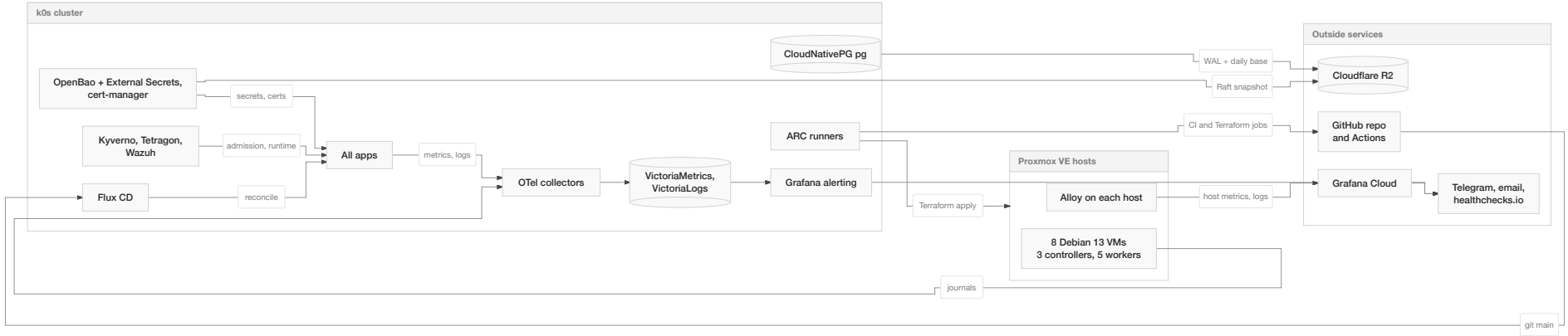
Edge. Cloudflare serves DNS, the tunnel and an offline page from a Worker while the lab sleeps (23:30 to 06:00).

Gateways. Cilium Gateway API runs `lab-gateway` (lab CA, HTTPS) and `public-gateway`. Pod traffic is WireGuard-encrypted.

Egress. Cilium FQDN policies limit agent runners to allowlisted hosts such as GitHub and the Anthropic API.

Control, telemetry and backups

Git is the only way to change the lab. Telemetry and backups leave the cluster, so an outage of the lab does not also remove the evidence or the data.



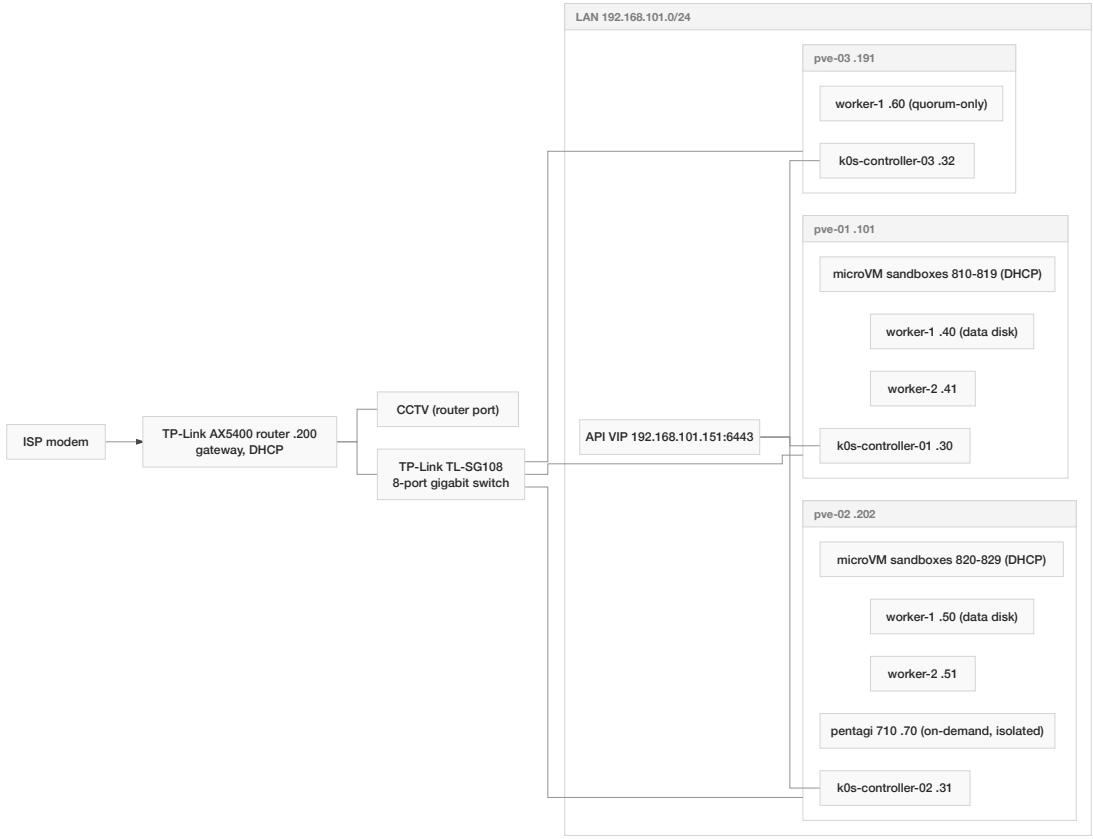
Change. PR, merge, Flux reconciles. Hand changes get reverted. Terraform for VMs runs on in-cluster ARC runners.

Observe. OTel collectors feed VictoriaMetrics and VictoriaLogs. Grafana alerts go to Telegram and email, with a healthchecks.io watchdog.

Recover. Postgres WAL and daily base backups (14 days PITR), OpenBao Raft snapshots and encrypted `k0s backup` go to R2.

Hardware and VMs

Three Proxmox VE hosts on a dedicated gigabit switch, so etcd, WireGuard and Postgres replication never cross the router. One controller per host keeps etcd quorum through a host failure.



Control plane. Three k0s controllers behind a keepalived VIP on `:6443`.

Placement. Stateful pods run only on workers labelled `lab/data-disk=true`. The pve-03 worker is quorum-only.

Sandboxes. microVM sandboxes and an isolated on-demand PenTAGI VM that tests only the public demo app.

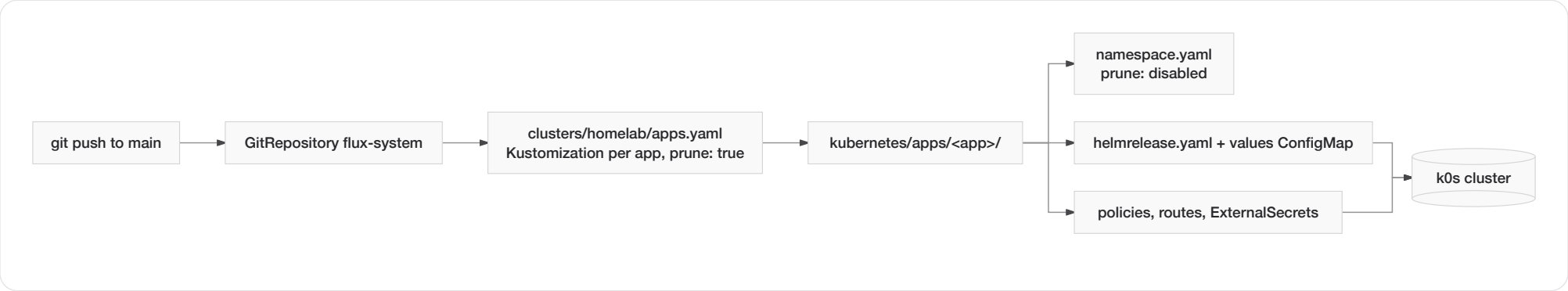
Platform services

What runs in the cluster, by concern.

CONCERN	COMPONENTS
Networking	Cilium CNI, WireGuard pod encryption, L2 announcements, LB IP pool, Gateway API
DNS and TLS	Pi-hole (two replicas) for `*.franzramadhan.lab`, external-dns for Cloudflare, cert-manager with Let's Encrypt (DNS-01) and a lab CA
Identity	Cloudflare Access (Google) as OIDC issuer for kube-apiserver, Proxmox, OpenBao, Headlamp, LiteLLM, and n8n through oauth2-proxy
Secrets	OpenBao (HA Raft) feeds External Secrets. Kubernetes Secrets encrypted in etcd (secretbox)
Policy and security	Kyverno (PSS, Flux-drift and hygiene policies), Tetragon tracing and kill policies, Wazuh, kube-apiserver audit log
Storage and backups	local-path-provisioner, CloudNativePG (2 instances), backups to Cloudflare R2 with tested restores
Observability	OpenTelemetry collectors, VictoriaMetrics, VictoriaLogs, Grafana with alert rules, VM journals over `systemd-journal-upload`, Alloy on hosts to Grafana Cloud
Edge protection	dvarapala: my own Rust rate limiter and WAF with a proof-of-work login challenge, one gateway per public app, managed by its operator
Wasm	runwasi Wasmtime shim and RuntimeClass `wasmtime`. ipcalc (`wasi:http`) and antree run as Wasm
AI and automation	LiteLLM gateway (Cloudflare Workers AI, smart routing, spend tracking), n8n with Claude Code runners for code review and PRs
Apps	Homie dashboard and a public demo, Headlamp with the Flux plugin, socialbtw (Threads reply moderation)
CI	GitHub Actions on an in-cluster ARC scale set (0 to 3 runners), Terraform plan and apply, Renovate for version bumps

GitOps and access zones

From bare metal to a running app with five repeatable steps, then one Flux Kustomization per app.



Provisioning

1. A script builds the Debian 13 template and pushes the cloud-init snippet to every host.
2. Terraform (bpg/proxmox) clones the VMs. `vms.tf` is the source of truth.
3. `k0sctl apply` installs k0s.
4. `kubectl apply -k` bootstraps Flux.
5. Flux reconciles every app from `main`.

Access zones

ZONE	REACH
Tailnet	Tailscale operator proxies the API server and the private UIs
LAN	Direct API, `lab-gateway`, Pi-hole, microVM sandboxes
Internet	Cloudflare Tunnel to `public-gateway`, only for `exposure=public` namespaces